Generating and Characteristic Functions

February 4, 2026

Generating and characteristic functions

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Probability generating function

Definition

Let X be a nonnegative integer-valued random variable. The probability generating function of X is defined to be

$$G_X(s) \equiv \mathbb{E}(s^X) = \sum_{k \geqslant 0} s^k \, \mathbb{P}(X = k)$$

For each fixed $s \in \mathbb{R}$, $G_X(s)$ is the expectation of the transformed random variable

$$g(X) = s^X$$

Probability generating function

- If X takes a finite number of values, $G_X(s)$ is a polynomial in the indeterminate s.

$$G_X(s) = \mathbb{P}(X = 0) + \mathbb{P}(X = 1)\, s + \cdots + \mathbb{P}(X = n)\, s^n$$

- Otherwise, if X takes a countable number of values, then $G_X(s)$ is expressed as a power series:

$$\begin{aligned} G_X(s) &= \sum_{k \geqslant 0} s^k\, \mathbb{P}(X = k) \\ &= \mathbb{P}(X = 0) + \mathbb{P}(X = 1)\, s + \cdots + \mathbb{P}(X = k)\, s^k + \cdots \end{aligned}$$

Probability generating function

The series

$$\sum_{k \geqslant 0} s^k \, \mathbb{P}(X = k)$$

converges at least for all $s \in [-1, 1]$, because if $|s| \leqslant 1$, then

$$\sum_{k \geqslant 0} |s|^k \, \mathbb{P}(X = k) \leqslant \sum_{k \geqslant 0} \mathbb{P}(X = k) = 1$$

Proposition

There exists a radius of convergence R, $1 \leqslant R \leqslant \infty$, such that $\sum_{k \geqslant 0} s^k \, \mathbb{P}(X = k)$ converges absolutely if $|s| < R$ and diverges if $|s| > R$.

Therefore, the probability generating function $G_X(s)$ is well-defined for all $s \in (-R, R)$.

Examples

Let X be a Bernoulli random variable, $X \sim \mathrm{Be}(p)$, such that

$$\mathbb{P}(X = 0) = q, \quad \mathbb{P}(X = 1) = p,$$

where $q = 1 - p$.

Then

$$G_X(s) = \sum_{k \geqslant 0} s^k \, \mathbb{P}(X = k) = q + ps, \quad s \in \mathbb{R}$$

Examples

If $X \sim \mathrm{Bin}(n, p)$, then

$$\mathbb{P}(X = k) = \binom{n}{k} p^k q^{n-k}, \quad k = 0, 1, \ldots, n$$

Thus

$$G_X(s) = \sum_{k \geqslant 0} s^k \, \mathbb{P}(X = k) = \sum_{k=0}^{n} s^k \binom{n}{k} p^k q^{n-k}$$

$$= \sum_{k=0}^{n} \binom{n}{k} (ps)^k q^{n-k} = (q + ps)^n, \quad s \in \mathbb{R}$$

Examples

Let $X \sim \mathrm{Po}(\lambda)$.

$$\mathbb{P}(X = k) = e^{-\lambda}\,\frac{\lambda^k}{k!}, \quad k = 0, 1, 2 \ldots$$

Then

$$\begin{aligned} G_X(s) &= \sum_{k \geqslant 0} s^k\,\mathbb{P}(X = k) \\ &= e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda s)^k}{k!} = e^{-\lambda} e^{\lambda s} = e^{\lambda(s-1)}, \quad s \in \mathbb{R} \end{aligned}$$

Examples

Let $X \sim \mathrm{Ge}(p)$.

$$\mathbb{P}(X = k) = q^{k-1}p, \quad k = 1, 2, \ldots, \quad 0 < p < 1$$

Therefore

$$G_X(s) = \sum_{k \geqslant 0} s^k \, \mathbb{P}(X = k) = \sum_{k=1}^{\infty} s^k q^{k-1} p$$

$$= ps \sum_{k=1}^{\infty} (qs)^{k-1} = \frac{ps}{1 - qs}, \quad |s| < \frac{1}{q}$$

Uniqueness

Standard results on power series establish that for all $k \geqslant 0$:

$$\mathbb{P}(X = k) = \frac{G_X^{(k)}(0)}{k!}$$

Therefore, the probability generating function uniquely specifies the probability distribution of X.

Theorem (uniqueness)

Let X and Y be nonnegative integer-valued random variables such that $G_X(s) = G_Y(s)$. Then

$$\mathbb{P}(X = k) = \mathbb{P}(Y = k) \quad \textit{for all } k \geqslant 0.$$

Convolution theorem

Theorem (convolution)

Let X and Y be independent, nonnegative integer-valued random variables, and let $Z = X + Y$. Then

$$G_Z(s) = G_X(s)G_Y(s)$$

Proof: Since X and Y are independent, the random variables s^X and s^Y are also independent. Therefore,

$$\begin{aligned} G_Z(s) &= \mathbb{E}\left(s^Z\right) = \mathbb{E}\left(s^{X+Y}\right) \\ &= \mathbb{E}\left(s^X s^Y\right) = \mathbb{E}\left(s^X\right)\mathbb{E}\left(s^Y\right) = G_X(s)G_Y(s) \end{aligned}$$

Convolution theorem

Example

Let $X \sim \text{Bin}(n, p)$ and $Y \sim \text{Bin}(m, p)$ be independent, and let

$$Z = X + Y$$

We have

$$\begin{aligned} G_Z(s) &= G_X(s) G_Y(s) \\ &= (q + ps)^n (q + ps)^m = (q + ps)^{n+m} \end{aligned}$$

By applying the uniqueness theorem, it follows that

$$X + Y \sim \text{Bin}(n + m, p)$$

Convolution theorem

More generally,

Theorem

Let $X_1, X_2, \ldots, X_n$ be independent, nonnegative integer-valued random variables, and set

$$Y = X_1 + X_2 + \cdots + X_n$$

Then

$$G_Y(s) = G_{X_1}(s) G_{X_2}(s) \cdots G_{X_n}(s).$$

Convolution theorem

An especially important case is as follows:

Corollary

If, in addition, $X_1, X_2, \ldots, X_n$ are equally distributed, with common probability generating function $G_X(s)$, then $G_Y(s) = (G_X(s))^n$.

Example

Let $X_1, X_2, \ldots, X_n$ be independent $\mathrm{Be}(p)$-distributed random variables and let $Y = X_1 + X_2 + \cdots + X_n$. Then

$$G_Y(s) = (G_X(s))^n = (q + sp)^n,$$

and so $Y \sim \mathrm{Bin}(n, p)$.

Example: Negative binomial distribution

Consider a coin with probability p of landing on heads, which is flipped repeatedly until a total of k heads is obtained.

Define X as the total number of tosses. Observe that

$$X = X_1 + X_2 + \cdots + X_k,$$

where X_i denotes the number of tosses between the $(i-1)$-th and the i-th occurrence of heads. Therefore, the variables $X_1, X_2, \ldots, X_k$ are independent and

$$X_i \sim \mathrm{Ge}(p), \quad 1 \leqslant i \leqslant k$$

Example: Negative binomial distribution

By the convolution theorem, we have

$$G_X(s) = (G_{X_1}(s))^k = \left(\frac{ps}{1-qs}\right)^k, \quad |s| < \frac{1}{q}$$

Expanding $G_X(s)$ as a power series

$$G_X(s) = \sum_{k \geqslant 0} a_k s^k$$

we get

$$a_k = \mathbb{P}(X = k), \quad k \geqslant 0$$

Example: Negative binomial distribution

The Taylor series expansion of the function $(1+x)^\alpha$ about zero, where α is a real number, is expressed as follows:

$$
\begin{aligned}
(1+x)^\alpha &= 1 + \alpha\, x + \frac{\alpha(\alpha-1)}{2}\, x^2 + \cdots \\
&\quad + \frac{\alpha(\alpha-1)\ldots(\alpha-r+1)}{r!}\, x^r + \cdots \\
&= \sum_{r \geqslant 0} \binom{\alpha}{r} x^r, \quad x \in (-1, 1),
\end{aligned}
$$

where $\binom{\alpha}{0} \equiv 1$ and, for $r \geqslant 1$,

$$
\binom{\alpha}{r} \equiv \frac{\alpha(\alpha-1)\cdots(\alpha-r+1)}{r!}
$$

Example: Negative binomial distribution

Identifying x with $-sq$ and α with $-k$ yields the following result:

$$G_X(s) = (ps)^k(1-qs)^{-k} = (ps)^k \sum_{r=0}^{\infty} \binom{-k}{r} (-qs)^r,$$

where

$$\begin{aligned}\binom{-k}{r} &= \frac{-k(-k-1)\cdots(-k-r+1)}{r!} \\ &= (-1)^r \binom{k+r-1}{k-1}, \quad r \geqslant 0\end{aligned}$$

Example: Negative binomial distribution

Therefore,

$$G_X(s) = \sum_{r=0}^{\infty} \binom{k+r-1}{k-1} p^k q^r s^{k+r} = \sum_{n=k}^{\infty} \binom{n-1}{k-1} p^k q^{n-k} s^n$$

Hence,

$$\mathbb{P}(X = n) = \begin{cases} 0, & n < k \\ \binom{n-1}{k-1} p^k q^{n-k}, & n = k, k+1, \cdots \end{cases}$$

This is the negative binomial probability law, $X \sim \mathrm{NBin}(k, p)$.

Additional properties

- $G_X(0) = \mathbb{P}(X = 0)$
- More generally, $\mathbb{P}(X = k) = \dfrac{G_X^{(k)}(0)}{k!}, \quad k \geqslant 0$
- $G_X(1) = 1$

  Indeed,

$$G_X(1) = \left[\sum_{k \geqslant 0} s^k \mathbb{P}(X = k)\right]_{s=1} = \sum_{k \geqslant 0} \mathbb{P}(X = k) = 1$$

Derivatives and moments

Proposition

Let R be the radius of covergence of $G_X(s)$. If $R > 1$, then

$$\mathbb{E}(X) = G'_X(1)$$

If we differentiate the series one term at a time, we get:

$$G'_X(s) = \frac{d}{ds} \sum_{k \geqslant 0} s^k \mathbb{P}(X = k) = \sum_{k \geqslant 1} k s^{k-1} \mathbb{P}(X = k)$$

Hence,

$$G'_X(1) = \sum_{k \geqslant 0} k \, \mathbb{P}(X = k) = \mathbb{E}(X)$$

Derivatives and moments

More generally,

Proposition

(a) $\mathbb{E}(X) = G'_X(1) \equiv \lim_{s \to 1^-} G'_X(s)$

(b) $\mathbb{E}(X(X-1)\cdots(X-k+1)) = G_X^{(k)}(1) \equiv \lim_{s \to 1^-} G_X^{(k)}(s)$

In the case $R = 1$, the value $\lim_{s \to 1^-} G_X^{(k)}(s)$ could be ∞.

Examples

Let $X \sim \text{Bin}(n, p)$.

$$\mathbb{E}(X) = G'_X(1) = \left.\frac{d}{ds}(q + ps)^n\right|_{s=1}$$

$$= \left.np\,(q + ps)^{n-1}\right|_{s=1} = np\,(q + p)^{n-1} = np$$

Examples

Let $X \sim \mathrm{Po}(\lambda)$.

$$\mathbb{E}(X) = G'_X(1) = \frac{d}{ds}\, e^{\lambda(s-1)}\Big|_{s=1} = \lambda\, e^{\lambda(s-1)}\Big|_{s=1} = \lambda$$

Analogously,

$$\mathbb{E}(X(X-1)) = G''_X(1) = \lambda^2\, e^{\lambda(s-1)}\Big|_{s=1} = \lambda^2$$

Hence, $\mathbb{E}(X^2) = \lambda^2 + \lambda$, and

$$\mathrm{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = \lambda$$

Examples

If $X \sim \mathrm{Ge}(p)$, then

$$\mathbb{E}(X) = G_X'(1) = \frac{d}{ds}\frac{ps}{(1-qs)}\bigg|_{s=1} = \frac{p}{(1-qs)^2}\bigg|_{s=1} = \frac{1}{p}$$

Analogously,

$$\mathbb{E}(X(X-1)) = G_X''(1) = \frac{2pq}{(1-qs)^3}\bigg|_{s=1} = \frac{2q}{p^2}$$

Therefore

$$E(X^2) = \frac{2q}{p^2} + \frac{1}{p}, \qquad \mathrm{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = \frac{q}{p^2}$$

Examples

$X \sim \text{NBin}(k, p)$.

$$\mathbb{E}(X) = G_X'(1) = \frac{d}{ds}\left(\frac{ps}{1-qs}\right)^k\Bigg|_{s=1}$$

$$= k\left(\frac{ps}{1-qs}\right)^{k-1}\frac{p}{(1-qs)^2}\Bigg|_{s=1} = \frac{k}{p}$$

You can also get this result by letting $X = \sum_{i=1}^k X_i$, where each $X_i \sim \text{Ge}(p)$. In this case,

$$\mathbb{E}(X) = \sum_{i=1}^k \mathbb{E}(X_i) = \frac{k}{p}.$$

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Moment generating function

Definition

The moment generating function of a random variable X is defined as

$$\Phi_X(t) \equiv \mathbb{E}\left(e^{tX}\right).$$

This definition holds for all values $t \in \mathbb{R}$ for which the above expectation exists.

- If X takes nonnegative integer values, then

$$\Phi_X(t) = \mathbb{E}\left(e^{tX}\right) = [G_X(s)]_{s=e^t} = G_X(e^t).$$

Moment generating function

For more general random variables we have

$$\Phi_X(t) = \begin{cases} \sum_i e^{tx_i}\, \mathbb{P}(X = x_i), & \textit{if } X \textit{ is discrete} \\ \int_{-\infty}^{\infty} e^{tx}\, f_X(x)\, dx, & \textit{if } X \textit{ is continuous} \end{cases},$$

provided that the sum or the integral converges.

For a continuous random variable, $\Phi_X(t)$ corresponds to the Laplace transform of the density function $f_X(x)$. (Notice the change of sign from the conventional definition of the Laplace transform.)

Examples

Let $X \sim \mathrm{Bin}(n, p)$.

$$\begin{aligned}\Phi_X(t) &= \sum_{k=0}^{n} e^{tk}\, \mathbb{P}(X = k) \\ &= \sum_{k=0}^{n} \binom{n}{k} \left(pe^t\right)^k q^{n-k} = \left(q + pe^t\right)^n, \quad t \in \mathbb{R}\end{aligned}$$

Examples

Let $X \sim \mathrm{Po}(\lambda)$.

$$\Phi_X(t) = \sum_{k=0}^{\infty} e^{tk}\, \mathbb{P}(X = k) = \sum_{k=0}^{\infty} e^{tk}\, e^{-\lambda}\, \frac{\lambda^k}{k!}$$

$$= e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda e^t)^k}{k!} = e^{\lambda(e^t - 1)}, \quad t \in \mathbb{R}$$

- Notice that in these two examples, $\Phi_X(t) = G_X(e^t)$.

Examples

Let $X \sim \text{Exp}(\mu)$.

$$\begin{aligned}\Phi_X(t) &= \int_{-\infty}^{\infty} e^{tx} f_X(x)\, dx \\ &= \int_0^{\infty} \mu e^{-(\mu-t)x}\, dx = \frac{\mu}{\mu - t}, \quad t < \mu\end{aligned}$$

Examples

Let $Z \sim \mathrm{N}(0,1)$. Then

$$\begin{aligned}
\Phi_Z(t) &= \int_{-\infty}^{\infty} e^{tz} f_Z(z)\, dz \\
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{z^2-2tz}{2}}\, dz \\
&= e^{\frac{t^2}{2}} \underbrace{\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{(z-t)^2}{2}}\, dz}_{1} \\
&= e^{t^2/2}, \quad t \in \mathbb{R}
\end{aligned}$$

Examples

More generally, let $Z \sim \mathrm{N}(0,1)$ and let $X = \sigma Z + m$, where $\sigma > 0$ and $m \in \mathbb{R}$.

Then $X \sim \mathrm{N}(m, \sigma^2)$ and

$$\begin{aligned}\Phi_X(t) &= \mathbb{E}\left(e^{tX}\right) = \mathbb{E}\left(e^{t(\sigma Z + m)}\right) \\ &= e^{tm}\,\mathbb{E}\left(e^{t\sigma Z}\right) = e^{tm}\,\Phi_Z(\sigma t) \\ &= \exp\left(\frac{\sigma^2 t^2}{2} + tm\right)\end{aligned}$$

Examples

Let X be a Cauchy random variable such that

$$f_X(x) = \frac{1/\pi}{1+x^2}, \quad -\infty < x < \infty$$

Then

$$\phi_X(t) = \int_{-\infty}^{\infty} e^{tx} \frac{1/\pi}{1+x^2}\, dx = \begin{cases} 1, & t = 0 \\ \infty & t \neq 0 \end{cases}$$

Therefore, in this example $\phi_X(t)$ exists only at $t = 0$.

Uniqueness

The moment generating function specifies uniquely the probability distribution.

Theorem

Let X and Y be random variables. If there exists $h > 0$ such that

$$\Phi_X(t) = \Phi_Y(t) \quad \text{for } |t| < h,$$

then X and Y are identically distributed.

Convolution theorem

Convolution also applies to moment generating functions.

Theorem

Let $X_1, X_2, \ldots, X_n$ be independent random variables and let

$$S = X_1 + X_2 + \cdots + X_n$$

Then,

$$\Phi_S(t) = \Phi_{X_1}(t)\Phi_{X_2}(t)\cdots\Phi_{X_n}(t)$$

Examples

Let $X \sim \text{Po}(\lambda_X)$ and $Y \sim \text{Po}(\lambda_Y)$ be independent.

If $Z = X + Y$, then

$$\begin{aligned}\Phi_Z(t) &= \Phi_X(t)\Phi_Y(t) \\ &= e^{\lambda_X(e^t-1)}\, e^{\lambda_Y(e^t-1)} \\ &= e^{(\lambda_X+\lambda_Y)(e^t-1)}\end{aligned}$$

Hence,

$$Z \sim \text{Po}(\lambda_X + \lambda_Y)$$

Examples

Let $X \sim \mathrm{N}(m_X, \sigma_X^2)$ and $Y \sim \mathrm{N}(m_Y, \sigma_Y^2)$ be independent.

If $Z = X + Y$, then

$$\begin{aligned}\Phi_Z(t) &= \Phi_X(t)\Phi_Y(t) \\ &= \exp\left(\frac{\sigma_X^2 t^2}{2} + t m_X\right) \exp\left(\frac{\sigma_Y^2 t^2}{2} + t m_Y\right) \\ &= \exp\left(\frac{(\sigma_X^2 + \sigma_Y^2) t^2}{2} + t(m_X + m_Y)\right)\end{aligned}$$

Therefore

$$Z \sim \mathrm{N}(m_X + m_Y, \sigma_X^2 + \sigma_Y^2)$$

Power series expansion

The following informal argument shows that

$$\Phi'_X(t) = \frac{d}{dt}\mathbb{E}\left(e^{tX}\right) = \mathbb{E}\left(\frac{d}{dt}e^{tX}\right) = \mathbb{E}\left(X\,e^{tX}\right)$$

Consequently, it follows that

$$\Phi'_X(0) = \mathbb{E}(X)$$

Analogously,

$$\Phi''_X(t) = \frac{d}{dt}\Phi'_X(t) = \frac{d}{dt}\mathbb{E}\left(X\,e^{tX}\right) = \mathbb{E}\left(X^2 e^{tX}\right)$$

and so

$$\Phi''_X(0) = \mathbb{E}(X^2)$$

Power series expansion

For instance, if $X \sim \text{Exp}(\mu)$, then

$$\mathbb{E}(X) = \Phi'_X(0) = \frac{d}{dt}\left(\frac{\mu}{\mu - t}\right)\bigg|_{t=0} = \frac{\mu}{(\mu - t)^2}\bigg|_{t=0} = \frac{1}{\mu}$$

Analogously,

$$\mathbb{E}(X^2) = \Phi''_X(0) = \frac{d}{dt}\left(\frac{\mu}{(\mu - t)^2}\right)\bigg|_{t=0} = \frac{2\mu}{(\mu - t)^3}\bigg|_{t=0} = \frac{2}{\mu^2}$$

Power series expansion

More generally,

$$\begin{aligned}\Phi_X(t) &= \mathbb{E}\left(e^{tX}\right)\\ &= \mathbb{E}\left(1 + t\,X + \frac{(t\,X)^2}{2!} + \cdots + \frac{(t\,X)^k}{k!} + \cdots\right)\\ &= 1 + \mathbb{E}(X)\,t + \frac{\mathbb{E}(X^2)}{2!}\,t^2 + \cdots + \frac{\mathbb{E}(X^k)}{k!}\,t^k + \cdots\end{aligned}$$

This is the Taylor's series expansion about zero of $\Phi_X(t)$, $\Phi_X(t) = \sum_{k=0}^{\infty} \Phi_X^{(k)}(0)/k!\,t^k$. Hence,

$$\mathbb{E}\left(X^k\right) = \Phi_X^{(k)}(0)$$

Moments

Theorem

If $\Phi_X(t)$ converges on an open interval containing $t = 0$, then the random variable X possesses moments of all orders, and

$$\mathbb{E}\left(X^k\right) = \Phi_X^{(k)}(0).$$

Furthermore,

$$\Phi_X(t) = \sum_{k=0}^{\infty} \frac{\mathbb{E}(X^k)}{k!}\, t^k$$

Examples

For instance, let $X \sim \mathrm{Exp}(\mu)$. If $|t| < \mu$, then

$$\Phi_X(t) = \frac{\mu}{\mu - t} = \frac{1}{1 - (t/\mu)}$$

$$= 1 + \frac{t}{\mu} + \left(\frac{t}{\mu}\right)^2 + \cdots + \left(\frac{t}{\mu}\right)^n + \cdots$$

We conclude that

$$\frac{\mathbb{E}\left(X^n\right)}{n!} = \frac{1}{\mu^n},$$

that is to say,

$$\mathbb{E}\left(X^n\right) = \frac{n!}{\mu^n}$$

Examples

Let $Z \sim \mathrm{N}(0,1)$. Then, for any $t \in \mathbb{R}$,

$$\Phi_Z(t) = e^{t^2/2} = 1 + \frac{t^2}{2} + \frac{t^4}{4 \cdot 2!} + \cdots + \frac{t^{2k}}{2^k \cdot k!} + \cdots$$

We conclude that $\mathbb{E}(Z^n) = 0$ if n is odd, and if $n = 2k$, then

$$\mathbb{E}\left(Z^{2k}\right) = \frac{2k!}{2^k \cdot k!} = (2k-1)!!,$$

where $(2k-1)!!$ denotes the product of all odd numbers from 1 to $2k-1$.

Uniqueness theorem for moments

Theorem

Suppose that all moments $\mathbb{E}(X)$, $\mathbb{E}(X^2)$, ... *of the random variable* X *exist, and that the series*

$$\sum_{k=0}^{\infty} \frac{\mathbb{E}(X^k)}{k!} t^k$$

is absolutely convergent for some $t > 0$. *Then the sequence of moments uniquely determines the distribution of* X.

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Characteristic function

Definition

The characteristic function $M_X(\omega)$ of a random variable X is the complex-valued function of the real argument ω defined as

$$M_X(\omega) \equiv \mathbb{E}\left(e^{i\,\omega X}\right)$$

- The expression $e^{i\omega X}$ represents a complex-valued random variable dependent on X, where $e^{i\omega X} \equiv \cos(\omega X) + i \sin(\omega X)$. Therefore,

$$M_X(\omega) = \mathbb{E}\left(\cos(\omega X)\right) + i\,\mathbb{E}\left(\sin(\omega X)\right)$$

Characteristic function

If X is a discrete or a continuous random variable, then

$$M_X(\omega) = \begin{cases} \sum_k e^{i\,\omega x_k}\,\mathbb{P}(X = x_k), & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} e^{i\,\omega x}\, f_X(x)\, dx, & \text{if } X \text{ is continuous} \end{cases}$$

- For a continuous random variable, $M_X(\omega)$ is essentially the Fourier transform of its probability density function $f_X(x)$. For discrete random variables, characteristic functions are closely related to Fourier series.

Properties

Proposition

Each random variable possesses a characteristic function. Furthermore, $M_X(\omega)$ is defined for all $\omega \in \mathbb{R}$.

- $|M_X(\omega)| \leqslant M_X(0) = 1$ for all $\omega \in \mathbb{R}$.

$$M_X(0) = \mathbb{E}\left(e^{i \cdot 0 \cdot X}\right) = \mathbb{E}(1) = 1$$

$$|M_X(\omega)| = \left|\mathbb{E}\left(e^{i\,\omega X}\right)\right| \leqslant \mathbb{E}\left(\left|e^{i\,\omega X}\right|\right) = \mathbb{E}(1) = 1$$

- The function $M_X(\omega)$ is uniformly continuous on the entire set $\mathbb{R}$.

Properties

- $\overline{M_X(\omega)} = M_X(-\omega)$.

$$\begin{aligned}
\overline{M_X(\omega)} &= \overline{\mathbb{E}\left(\cos\left(\omega X\right)\right) + i\,\mathbb{E}\left(\sin\left(\omega X\right)\right)} \\
&= \mathbb{E}\left(\cos\left(\omega X\right)\right) - i\,\mathbb{E}\left(\sin\left(\omega X\right)\right) \\
&= \mathbb{E}\left(\cos\left(-\omega X\right)\right) + i\,\mathbb{E}\left(\sin\left(-\omega X\right)\right) \\
&= M_X(-\omega)
\end{aligned}$$

More concisely,

$$\begin{aligned}
\overline{M_X(\omega)} &= \overline{\mathbb{E}\left(e^{i\,\omega X}\right)} = \mathbb{E}\left(\overline{e^{i\,\omega X}}\right) \\
&= \mathbb{E}\left(e^{-i\,\omega X}\right) = \mathbb{E}\left(e^{i\,(-\omega)X}\right) \\
&= M_X(-\omega)
\end{aligned}$$

Examples

- Let $X \sim \mathrm{Bin}(n, p)$. Then, $M_X(\omega) = \left(q + p\, e^{i\omega}\right)^n$.
- If $X \sim \mathrm{Po}(\lambda)$, then $M_X(\omega) = e^{\lambda\left(e^{i\omega}-1\right)}$.
- If $X \sim \mathrm{N}(0, 1)$, then $M_X(\omega) = e^{-\frac{1}{2}\omega^2}$.
- If X has the Cauchy distribution, then

$$M_X(\omega) = \int_{-\infty}^{\infty} e^{i\omega x} \frac{1/\pi}{1 + x^2}\, dx = e^{-|\omega|},$$

  a result obtainable by the calculus of residues.

Examples

- If $Y = aX + b$, then

$$\begin{aligned} M_Y(\omega) &= \mathbb{E}\left(e^{i\omega(aX+b)}\right) \\ &= \mathbb{E}\left(e^{i\omega b} e^{i\omega aX}\right) = e^{i\omega b}\,\mathbb{E}\left(e^{i\omega aX}\right) \\ &= e^{i\omega b} M_X(a\,\omega) \end{aligned}$$

- For instance, if $Z \sim \mathrm{N}(0,1)$ and $X = \sigma Z + m$, $\sigma > 0$, then $X \sim \mathrm{N}(m, \sigma^2)$, and

$$M_X(\omega) = e^{i\omega m - \frac{1}{2}\sigma^2\omega^2}$$

Inversion

Theorem (inversion of the Fourier transform)

Let X be a continuous random variable with density $f_X(x)$ and characteristic function $M_X(\omega)$. Then

$$f_X(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega x} M_X(\omega)\, d\omega$$

at every point x at which $f_X(x)$ is differentiable.

- To obtain $f_X(x)$ from $M_X(\omega)$ usually requires contour integration in the complex plane.

Inversion

In the discrete case,

Theorem

If X is an integer-valued random variable, then

$$\mathbb{P}(X = k) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-ik\omega} M_X(\omega)\, d\omega$$

Inversion and uniqueness

Theorem

Let X have distribution function $F_X(x)$ and characteristic function $M_X(\omega)$. Let

$$\overline{F}_X(x) = \frac{F_X(x) + F_X(x^-)}{2}.$$

Then

$$\overline{F}_X(b) - \overline{F}_X(a) = \lim_{T\to\infty} \frac{1}{2\pi} \int_{-T}^{T} \frac{e^{-ia\omega} - e^{-ib\omega}}{i\omega} M_X(\omega)\, d\omega.$$

Therefore, the characteristic function $M_X(\omega)$ uniquely defines the probability law of the random variable X.

Convolution theorem

Theorem

Let $X_1, X_2, \ldots, X_n$ be independent random variables and let

$$Y = X_1 + X_2 + \cdots + X_n.$$

Then

$$M_Y(\omega) = M_{X_1}(\omega) M_{X_2}(\omega) \cdots M_{X_n}(\omega)$$

Convolution theorem

Proof:

$$
\begin{aligned}
M_Y(\omega) &= \mathbb{E}\left(e^{i\omega Y}\right) \\
&= \mathbb{E}\left(e^{i\omega\,(X_1+X_2+\cdots+X_n)}\right) \\
&= \mathbb{E}\left(e^{i\omega X_1} e^{i\omega X_2} \cdots e^{i\omega X_n}\right) \\
&= \mathbb{E}\left(e^{i\omega X_1}\right) \mathbb{E}\left(e^{i\omega X_2}\right) \cdots \mathbb{E}\left(e^{i\omega X_n}\right) \\
&= M_{X_1}(\omega) M_{X_2}(\omega) \cdots M_{X_n}(\omega)
\end{aligned}
$$

Convolution theorem

(Remark)

We essentially have the convolution theorem for Fourier transforms.

If X and Y are continuous and independent random variables and $Z = X + Y$, then $f_Z = f_X * f_Y$, which implies

$$\mathcal{F}(f_Z) = \mathcal{F}(f_X) \cdot \mathcal{F}(f_Y),$$

that is to say,

$$M_Z(\omega) = M_X(\omega) M_Y(\omega)$$

Characteristic function and moments

If the random variable X has moments of any order, then we have:

$$M_X(\omega) = \mathbb{E}\left(e^{i\omega X}\right)$$

$$= \mathbb{E}\left(\sum_{k=0}^{\infty} \frac{(i\,\omega X)^k}{k!}\right) = \sum_{k=0}^{\infty} \frac{i^k\,\mathbb{E}(X^k)}{k!}\,\omega^k$$

This is the Taylor series expansion for $M_X(\omega)$. We conclude that $i^k\mathbb{E}(X^k) = M_X^{(k)}(0)$, and hence

$$\mathbb{E}(X^k) = \frac{M_X^{(k)}(0)}{i^k}, \quad k \geqslant 1$$

Characteristic function and moments

More generally,

Theorem

If $\mathbb{E}(X^n) < \infty$ *for some* $n \geqslant 1$, *then*

$$M_X(\omega) = \sum_{k=0}^{n} \frac{\mathbb{E}(X^k)}{k!}(i\,\omega)^k + o\left(|\omega|^n\right) \quad \text{as} \quad \omega \to 0,$$

and

$$\mathbb{E}(X^k) = \frac{M_X^{(k)}(0)}{i^k} \quad \text{for} \quad k = 1, 2, \ldots, n.$$

Joint characteristic functions

Definition

The joint characteristic function of the random variables X_1, X_2, ..., X_n is defined to be

$$M_X(\omega_1, \omega_2, \ldots, \omega_n) \equiv \mathbb{E}\left(e^{i(\omega_1 X_1 + \omega_2 X_2 + \cdots + \omega_n X_n)}\right)$$

Using vectorial notation one can write $\boldsymbol{X} = (X_1, X_2, \ldots, X_n)^t$, $\boldsymbol{\omega} = (\omega_1, \omega_2, \ldots, \omega_n)^t$, and

$$M_X(\boldsymbol{\omega}^t) = \mathbb{E}\left(e^{i\boldsymbol{\omega}^t \cdot \boldsymbol{X}}\right)$$

Joint moments

Joint characteristic functions facilitate the calculation of joint moments.

For instance, given the joint characteristic function of the random variables X and Y,

$$M_{XY}(\omega_1, \omega_2) = \mathbb{E}\left(e^{i(\omega_1 X + \omega_2 Y)}\right),$$

we have

$$\frac{\partial M_{XY}(\omega_1, \omega_2)}{\partial \omega_1} = i\,\mathbb{E}\left(X\, e^{i(\omega_1 X + \omega_2 Y)}\right)$$

$$\frac{\partial^2 M_{XY}(\omega_1, \omega_2)}{\partial \omega_1 \partial \omega_2} = i^2\,\mathbb{E}\left(XY\, e^{i(\omega_1 X + \omega_2 Y)}\right)$$

Joint moments

Hence,

$$\left.\frac{\partial^2 M_{XY}(\omega_1,\omega_2)}{\partial\omega_1\partial\omega_2}\right|_{(\omega_1=0,\omega_2=0)} = i^2\,\mathbb{E}(XY) = -\mathbb{E}(XY)$$

In general, we have

For all $k \geqslant 0$ and $l \geqslant 0$,

$$\mathbb{E}\left(X^k Y^l\right) = \frac{1}{i^{k+l}}\ \left.\frac{\partial^{k+l} M_{XY}(\omega_1,\omega_2)}{\partial^k\omega_1\ \partial^l\omega_2}\right|_{(\omega_1,\omega_2)=(0,0)}$$

Marginal characteristic functions

Marginal characteristic functions are easily derived from the joint characteristic function.

For instance, given X and Y:

$$\begin{aligned} M_X(\omega) &= \mathbb{E}\left(e^{i\omega X}\right) \\ &= \mathbb{E}\left(e^{i(\omega_1 X + \omega_2 Y)}\right)\Big|_{(\omega_1=\omega,\, \omega_2=0)} = M_{XY}(\omega, 0) \end{aligned}$$

Analogously,

$$M_Y(\omega) = M_{XY}(0, \omega)$$

Independent random variables

Theorem

The random variables $X_1, X_2, \ldots, X_n$ are independent if and only if

$$M_X(\omega_1, \omega_2, \ldots, \omega_n) = M_{X_1}(\omega_1)\, M_{X_2}(\omega_2) \cdots M_{X_n}(\omega_n)$$

The necessity of the condition is easily proved.

$$\begin{aligned} M_X(\omega_1, \omega_2, \ldots, \omega_n) &= \mathbb{E}\left(e^{i(\omega_1 X_1 + \omega_2 X_2 + \cdots + \omega_n X_n)}\right) \\ &= \mathbb{E}\left(e^{i\omega_1 X_1}\, e^{i\omega_2 X_2}\, \cdots e^{i\omega_n X_n}\right) \\ &= \mathbb{E}\left(e^{i\omega_1 X_1}\right) \mathbb{E}\left(e^{i\omega_2 X_2}\right) \cdots \mathbb{E}\left(e^{i\omega_n X_n}\right) \\ &= M_{X_1}(\omega_1)\, M_{X_2}(\omega_2)\, \cdots\, M_{X_n}(\omega_n) \end{aligned}$$